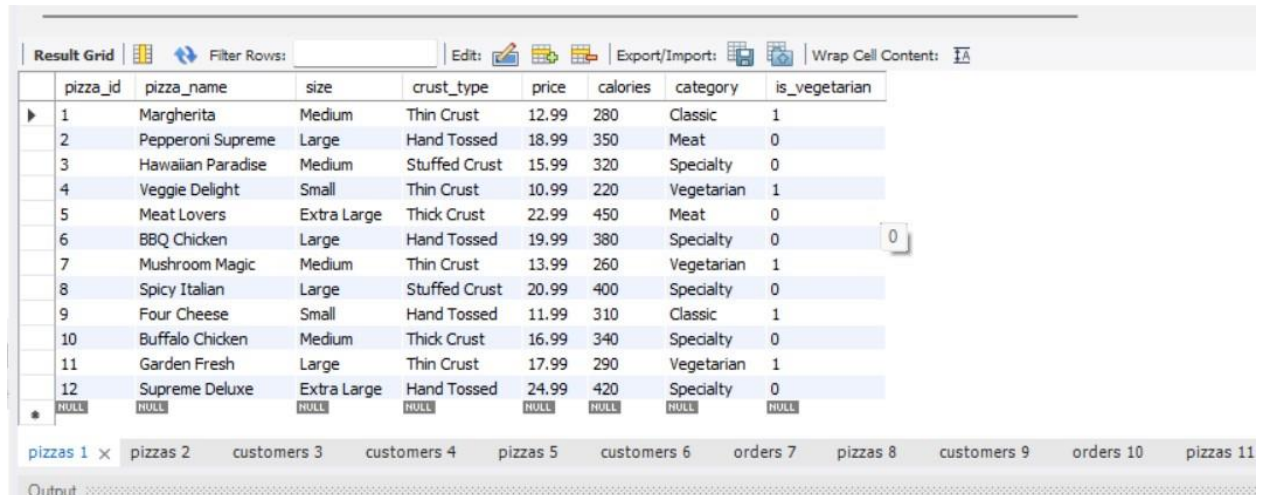


## SQL ASSIGNMENT-2

### Section A: Basic SELECT Statements

**Q1. Write a query to display all information about pizzas.**

SELECT \* FROM pizzas;



|   | pizza_id | pizza_name        | size        | crust_type    | price | calories | category   | is_vegetarian |
|---|----------|-------------------|-------------|---------------|-------|----------|------------|---------------|
| ▶ | 1        | Margherita        | Medium      | Thin Crust    | 12.99 | 280      | Classic    | 1             |
|   | 2        | Pepperoni Supreme | Large       | Hand Tossed   | 18.99 | 350      | Meat       | 0             |
|   | 3        | Hawaiian Paradise | Medium      | Stuffed Crust | 15.99 | 320      | Specialty  | 0             |
|   | 4        | Veggie Delight    | Small       | Thin Crust    | 10.99 | 220      | Vegetarian | 1             |
|   | 5        | Meat Lovers       | Extra Large | Thick Crust   | 22.99 | 450      | Meat       | 0             |
|   | 6        | BBQ Chicken       | Large       | Hand Tossed   | 19.99 | 380      | Specialty  | 0             |
|   | 7        | Mushroom Magic    | Medium      | Thin Crust    | 13.99 | 260      | Vegetarian | 1             |
|   | 8        | Spicy Italian     | Large       | Stuffed Crust | 20.99 | 400      | Specialty  | 0             |
|   | 9        | Four Cheese       | Small       | Hand Tossed   | 11.99 | 310      | Classic    | 1             |
|   | 10       | Buffalo Chicken   | Medium      | Thick Crust   | 16.99 | 340      | Specialty  | 0             |
|   | 11       | Garden Fresh      | Large       | Thin Crust    | 17.99 | 290      | Vegetarian | 1             |
|   | 12       | Supreme Deluxe    | Extra Large | Hand Tossed   | 24.99 | 420      | Specialty  | 0             |
| • | NULL     | NULL              | NULL        | NULL          | NULL  | NULL     | NULL       | NULL          |

**Q2. Display only the pizza\_name, size, and price from the pizzas table.**

SELECT pizza\_name, size, price FROM pizzas;



|   | pizza_name        | size        | price |
|---|-------------------|-------------|-------|
| ▶ | Margherita        | Medium      | 12.99 |
|   | Pepperoni Supreme | Large       | 18.99 |
|   | Hawaiian Paradise | Medium      | 15.99 |
|   | Veggie Delight    | Small       | 10.99 |
|   | Meat Lovers       | Extra Large | 22.99 |
|   | BBQ Chicken       | Large       | 19.99 |
|   | Mushroom Magic    | Medium      | 13.99 |
|   | Spicy Italian     | Large       | 20.99 |
|   | Four Cheese       | Small       | 11.99 |
|   | Buffalo Chicken   | Medium      | 16.99 |
|   | Garden Fresh      | Large       | 17.99 |
|   | Supreme Deluxe    | Extra Large | 24.99 |

**Q3. Show all customer information from the customer's table.**

SELECT \* FROM customers;

| Result Grid  |             |            |           |          |                          |               |            |                |          |
|--------------|-------------|------------|-----------|----------|--------------------------|---------------|------------|----------------|----------|
| Filter Rows: |             |            |           |          |                          |               |            |                |          |
|              | customer_id | first_name | last_name | phone    | email                    | city          | join_date  | loyalty_points | status   |
| ▶            | 201         | Tony       | Soprano   | 555-0101 | tony.soprano@email.com   | Newark        | 2023-01-15 | 450            | VIP      |
|              | 202         | Maria      | Rossi     | 555-0102 | maria.rossi@email.com    | Brooklyn      | 2023-02-20 | 280            | Regular  |
|              | 203         | Giuseppe   | Bianchi   | 555-0103 | giuseppe.b@email.com     | Manhattan     | 2022-11-10 | 750            | Gold     |
|              | 204         | Isabella   | Ferrari   | 555-0104 | isabella.f@email.com     | Queens        | 2023-03-05 | 120            | Regular  |
|              | 205         | Marco      | Lombardi  | 555-0105 | marco.lombardi@email.com | Bronx         | 2022-08-18 | 890            | Gold     |
|              | 206         | Sofia      | Romano    | 555-0106 | sofia.romano@email.com   | Brooklyn      | 2023-04-12 | 65             | Regular  |
|              | 207         | Antonio    | Conti     | 555-0107 | antonio.conti@email.com  | Staten Island | 2022-12-03 | 340            | VIP      |
|              | 208         | Francesca  | Moretti   | 555-0108 | francesca.m@email.com    | Manhattan     | 2023-01-28 | 180            | Regular  |
|              | 209         | Luca       | Esposito  | 555-0109 | luca.esposito@email.com  | Queens        | 2022-09-15 | 520            | VIP      |
|              | 210         | Giulia     | Ricci     | 555-0110 | giulia.ricci@email.com   | Bronx         | 2023-05-22 | 15             | Inactive |
|              |             | NULL       | NULL      | NULL     | NULL                     | NULL          | NULL       | NULL           | NULL     |

**Q4. Display only the first\_name, last\_name, and city of all customers**

SELECT first\_name, last\_name, city FROM customers;

| Result Grid  |            |           |               |
|--------------|------------|-----------|---------------|
| Filter Rows: |            |           |               |
|              | first_name | last_name | city          |
| ▶            | Tony       | Soprano   | Newark        |
|              | Maria      | Rossi     | Brooklyn      |
|              | Giuseppe   | Bianchi   | Manhattan     |
|              | Isabella   | Ferrari   | Queens        |
|              | Marco      | Lombardi  | Bronx         |
|              | Sofia      | Romano    | Brooklyn      |
|              | Antonio    | Conti     | Staten Island |
|              | Francesca  | Moretti   | Manhattan     |
|              | Luca       | Esposito  | Queens        |
|              | Giulia     | Ricci     | Bronx         |

## Section B: WHERE Clause

**Q5. Find all pizzas that are 'Large' size.**

SELECT \* FROM pizzas

WHERE size="Large";

|   | pizza_id | pizza_name        | size  | crust_type    | price | calories | category   | is_vegetarian |
|---|----------|-------------------|-------|---------------|-------|----------|------------|---------------|
| ▶ | 2        | Pepperoni Supreme | Large | Hand Tossed   | 18.99 | 350      | Meat       | 0             |
|   | 6        | BBQ Chicken       | Large | Hand Tossed   | 19.99 | 380      | Specialty  | 0             |
|   | 8        | Spicy Italian     | Large | Stuffed Crust | 20.99 | 400      | Specialty  | 0             |
|   | 11       | Garden Fresh      | Large | Thin Crust    | 17.99 | 290      | Vegetarian | 1             |
| * | NULL     | NULL              | NULL  | NULL          | NULL  | NULL     | NULL       | NULL          |

**Q6. Display all customers from 'Brooklyn'.**

SELECT \* FROM customers

WHERE city="Brooklyn";

|   | customer_id | first_name | last_name | phone    | email                  | city     | join_date  | loyalty_points | status  |
|---|-------------|------------|-----------|----------|------------------------|----------|------------|----------------|---------|
| ▶ | 202         | Maria      | Rossi     | 555-0102 | maria.rossi@email.com  | Brooklyn | 2023-02-20 | 280            | Regular |
|   | 206         | Sofia      | Romano    | 555-0106 | sofia.romano@email.com | Brooklyn | 2023-04-12 | 65             | Regular |
| * | NULL        | NULL       | NULL      | NULL     | NULL                   | NULL     | NULL       | NULL           | NULL    |

**Q7. Show all orders with total\_amount greater than \$40.**

SELECT \* FROM orders

WHERE total\_amount>40;

|   | order_id | customer_id | order_date | delivery_time | total_amount | delivery_type | order_status | special_instructions       |
|---|----------|-------------|------------|---------------|--------------|---------------|--------------|----------------------------|
| ▶ | 301      | 201         | 2024-01-15 | 18:30:00      | 45.97        | Delivery      | Delivered    | Extra cheese on all pizzas |
|   | 303      | 203         | 2024-01-16 | 19:45:00      | 67.95        | Dine-in       | Delivered    | NULL                       |
|   | 307      | 207         | 2024-01-18 | 19:00:00      | 41.97        | Dine-in       | Cancelled    | Table for 4                |
|   | 309      | 209         | 2024-01-19 | 21:30:00      | 52.96        | Delivery      | Delivered    | Ring doorbell twice        |
| * | NULL     | NULL        | NULL       | NULL          | NULL         | NULL          | NULL         | NULL                       |

**Q8. Find all vegetarian pizzas (is\_vegetarian = TRUE).**

SELECT \* FROM pizzas

WHERE is\_vegetarian=1;

127 WHERE order\_date="2024-01-17";

128

129 #Q11

Result Grid

|   | pizza_id | pizza_name     | size   | crust_type  | price | calories | category   | is_vegetarian |
|---|----------|----------------|--------|-------------|-------|----------|------------|---------------|
| ▶ | 1        | Margherita     | Medium | Thin Crust  | 12.99 | 280      | Classic    | 1             |
|   | 4        | Veggie Delight | Small  | Thin Crust  | 10.99 | 220      | Vegetarian | 1             |
|   | 7        | Mushroom Magic | Medium | Thin Crust  | 13.99 | 260      | Vegetarian | 1             |
|   | 9        | Four Cheese    | Small  | Hand Tossed | 11.99 | 310      | Classic    | 1             |
|   | 11       | Garden Fresh   | Large  | Thin Crust  | 17.99 | 290      | Vegetarian | 1             |
| * | NULL     | NULL           | NULL   | NULL        | NULL  | NULL     | NULL       | NULL          |

**Q9. Display all customers with 'VIP' status.**

SELECT \* FROM customers

WHERE status="VIP";

128

129 #Q11

Result Grid

|   | customer_id | first_name | last_name | phone    | email                   | city          | join_date  | loyalty_points | status |
|---|-------------|------------|-----------|----------|-------------------------|---------------|------------|----------------|--------|
| ▶ | 201         | Tony       | Soprano   | 555-0101 | tony.soprano@email.com  | Newark        | 2023-01-15 | 450            | VIP    |
|   | 207         | Antonio    | Conti     | 555-0107 | antonio.conti@email.com | Staten Island | 2022-12-03 | 340            | VIP    |
|   | 209         | Luca       | Esposito  | 555-0109 | luca.esposito@email.com | Queens        | 2022-09-15 | 520            | VIP    |
| * | NULL        | NULL       | NULL      | NULL     | NULL                    | NULL          | NULL       | NULL           | NULL   |

**Q10. Show all orders placed on '2024-01-17'**

SELECT \* FROM orders

WHERE order\_date="2024-01-17";

Result Grid

|   | order_id | customer_id | order_date | delivery_time | total_amount | delivery_type | order_status | special_instructions |
|---|----------|-------------|------------|---------------|--------------|---------------|--------------|----------------------|
| ▶ | 304      | 204         | 2024-01-17 | 20:00:00      | 32.99        | Delivery      | Preparing    | Call before delivery |
|   | 305      | 205         | 2024-01-17 | 13:30:00      | 18.99        | Pickup        | Ready        | Extra spicy          |
| * | NULL     | NULL        | NULL       | NULL          | NULL         | NULL          | NULL         | NULL                 |

## Section C: LIKE Operator

**Q11. Find all pizzas whose name starts with 'M'.**

```
SELECT * FROM pizzas
WHERE pizza_name LIKE "M%";
```

| Result Grid                       |          |                |             |             |       |          |            |               |
|-----------------------------------|----------|----------------|-------------|-------------|-------|----------|------------|---------------|
| Filter Rows: <input type="text"/> |          |                |             |             |       |          |            |               |
|                                   | pizza_id | pizza_name     | size        | crust_type  | price | calories | category   | is_vegetarian |
| ▶                                 | 1        | Margherita     | Medium      | Thin Crust  | 12.99 | 280      | Classic    | 1             |
|                                   | 5        | Meat Lovers    | Extra Large | Thick Crust | 22.99 | 450      | Meat       | 0             |
|                                   | 7        | Mushroom Magic | Medium      | Thin Crust  | 13.99 | 260      | Vegetarian | 1             |
| •                                 | NULL     | NULL           | NULL        | NULL        | NULL  | NULL     | NULL       | NULL          |

**Q12. Display all customers whose first name ends with 'a'.**

```
SELECT * FROM customers
WHERE first_name LIKE "%a";
```

|   | customer_id | first_name | last_name | phone    | email                   | city      | join_date  | loyalty_points | status   |
|---|-------------|------------|-----------|----------|-------------------------|-----------|------------|----------------|----------|
| ▶ | 202         | Maria      | Rossi     | 555-0102 | maria.rossi@email.com   | Brooklyn  | 2023-02-20 | 280            | Regular  |
|   | 204         | Isabella   | Ferrari   | 555-0104 | isabella.f@email.com    | Queens    | 2023-03-05 | 120            | Regular  |
|   | 206         | Sofia      | Romano    | 555-0106 | sofia.romano@email.com  | Brooklyn  | 2023-04-12 | 65             | Regular  |
|   | 208         | Francesca  | Moretti   | 555-0108 | francesca.m@email.com   | Manhattan | 2023-01-28 | 180            | Regular  |
|   | 209         | Luca       | Esposito  | 555-0109 | luca.esposito@email.com | Queens    | 2022-09-15 | 520            | VIP      |
|   | 210         | Giulia     | Ricci     | 555-0110 | giulia.ricci@email.com  | Bronx     | 2023-05-22 | 15             | Inactive |
| • | NULL        | NULL       | NULL      | NULL     | NULL                    | NULL      | NULL       | NULL           | NULL     |

**Q13. Show all pizzas that contain the word 'Chicken' in their name.**

```
SELECT * FROM pizzas
WHERE pizza_name LIKE "%Chicken%";
```

| Result Grid                       |          |                 |        |             |       |          |           |               |
|-----------------------------------|----------|-----------------|--------|-------------|-------|----------|-----------|---------------|
| Filter Rows: <input type="text"/> |          |                 |        |             |       |          |           |               |
|                                   | pizza_id | pizza_name      | size   | crust_type  | price | calories | category  | is_vegetarian |
| ▶                                 | 6        | BBQ Chicken     | Large  | Hand Tossed | 19.99 | 380      | Specialty | 0             |
|                                   | 10       | Buffalo Chicken | Medium | Thick Crust | 16.99 | 340      | Specialty | 0             |
| •                                 | NULL     | NULL            | NULL   | NULL        | NULL  | NULL     | NULL      | NULL          |

**Q14. Find all customers whose email contains 'gmail' (Note: modify some emails if needed for practice).**

```
SELECT * FROM customers
WHERE email LIKE "%gmail%";
```

| Result Grid                       |             |            |           |       |       |      |           |                |        |
|-----------------------------------|-------------|------------|-----------|-------|-------|------|-----------|----------------|--------|
| Filter Rows: <input type="text"/> |             |            |           |       |       |      |           |                |        |
|                                   | customer_id | first_name | last_name | phone | email | city | join_date | loyalty_points | status |
| •                                 | NULL        | NULL       | NULL      | NULL  | NULL  | NULL | NULL      | NULL           | NULL   |

**Q15. Display all pizzas whose name contains 'Supreme'.**

```
SELECT * FROM pizzas
```

WHERE pizza\_name LIKE "%Supreme%";

|   | pizza_id | pizza_name        | size        | crust_type  | price | calories | category  | is_vegetarian |
|---|----------|-------------------|-------------|-------------|-------|----------|-----------|---------------|
| ▶ | 2        | Pepperoni Supreme | Large       | Hand Tossed | 18.99 | 350      | Meat      | 0             |
|   | 12       | Supreme Deluxe    | Extra Large | Hand Tossed | 24.99 | 420      | Specialty | 0             |
| * | NULL     | NULL              | NULL        | NULL        | NULL  | NULL     | NULL      | NULL          |

## Section D: Comparison Operators

**Q16. Find all pizzas with price between \$15 and \$20.**

SELECT \* FROM pizzas

WHERE price BETWEEN 15 AND 20;

|   | pizza_id | pizza_name        | size   | crust_type    | price | calories | category   | is_vegetarian |
|---|----------|-------------------|--------|---------------|-------|----------|------------|---------------|
| ▶ | 2        | Pepperoni Supreme | Large  | Hand Tossed   | 18.99 | 350      | Meat       | 0             |
|   | 3        | Hawaiian Paradise | Medium | Stuffed Crust | 15.99 | 320      | Specialty  | 0             |
|   | 6        | BBQ Chicken       | Large  | Hand Tossed   | 19.99 | 380      | Specialty  | 0             |
|   | 10       | Buffalo Chicken   | Medium | Thick Crust   | 16.99 | 340      | Specialty  | 0             |
|   | 11       | Garden Fresh      | Large  | Thin Crust    | 17.99 | 290      | Vegetarian | 1             |
| * | NULL     | NULL              | NULL   | NULL          | NULL  | NULL     | NULL       | NULL          |

pizzas 17 customers 18 pizzas 19 orders 20 customers 21 pizzas 22 customers 23 pizzas 24

**Q17. Display customers with loyalty points greater than or equal to 500.**

SELECT \* FROM customers

WHERE loyalty\_points >= 500;

|   | customer_id | first_name | last_name | phone    | email                    | city      | join_date  | loyalty_points | status |
|---|-------------|------------|-----------|----------|--------------------------|-----------|------------|----------------|--------|
| ▶ | 203         | Giuseppe   | Bianchi   | 555-0103 | giuseppe.b@email.com     | Manhattan | 2022-11-10 | 750            | Gold   |
|   | 205         | Marco      | Lombardi  | 555-0105 | marco.lombardi@email.com | Bronx     | 2022-08-18 | 890            | Gold   |
|   | 209         | Luca       | Esposito  | 555-0109 | luca.esposito@email.com  | Queens    | 2022-09-15 | 520            | VIP    |
| * | NULL        | NULL       | NULL      | NULL     | NULL                     | NULL      | NULL       | NULL           | NULL   |

**Q18. Show all pizzas with calories less than 300.**

SELECT \* FROM pizzas

WHERE calories < 300;

165 • `SELECT * FROM pizzas`

|   | pizza_id | pizza_name     | size   | crust_type | price | calories | category   | is_vegetarian |
|---|----------|----------------|--------|------------|-------|----------|------------|---------------|
| ▶ | 1        | Margherita     | Medium | Thin Crust | 12.99 | 280      | Classic    | 1             |
|   | 4        | Veggie Delight | Small  | Thin Crust | 10.99 | 220      | Vegetarian | 1             |
|   | 7        | Mushroom Magic | Medium | Thin Crust | 13.99 | 260      | Vegetarian | 1             |
|   | 11       | Garden Fresh   | Large  | Thin Crust | 17.99 | 290      | Vegetarian | 1             |
| * | NULL     | NULL           | NULL   | NULL       | NULL  | NULL     | NULL       | NULL          |

**Q19. Find orders with total\_amount less than or equal to \$25.**

```
SELECT * FROM orders
WHERE total_amount <= 25;
```

|   | order_id | customer_id | order_date | delivery_time | total_amount | delivery_type | order_status | special_instructions |
|---|----------|-------------|------------|---------------|--------------|---------------|--------------|----------------------|
| ▶ | 302      | 202         | 2024-01-16 | 12:15:00      | 23.98        | Pickup        | Ready        | No onions please     |
|   | 305      | 205         | 2024-01-17 | 13:30:00      | 18.99        | Pickup        | Ready        | Extra spicy          |
|   | 308      | 208         | 2024-01-19 | 14:45:00      | 15.99        | Pickup        | Delivered    | Light sauce          |
|   | 310      | 210         | 2024-01-20 | 16:20:00      | 12.99        | Pickup        | Pending      | NULL                 |
| * | NULL     | NULL        | NULL       | NULL          | NULL         | NULL          | NULL         | NULL                 |

**Q20. Display all customers who joined before '2023-01-01'**

```
SELECT * FROM customers
WHERE join_date < "2023-01-01";
```

165 • `SELECT * FROM pizzas`

|   | customer_id | first_name | last_name | phone    | email                    | city          | join_date  | loyalty_points | status |
|---|-------------|------------|-----------|----------|--------------------------|---------------|------------|----------------|--------|
| ▶ | 203         | Giuseppe   | Bianchi   | 555-0103 | giuseppe.b@email.com     | Manhattan     | 2022-11-10 | 750            | Gold   |
|   | 205         | Marco      | Lombardi  | 555-0105 | marco.lombardi@email.com | Bronx         | 2022-08-18 | 890            | Gold   |
|   | 207         | Antonio    | Conti     | 555-0107 | antonio.conti@email.com  | Staten Island | 2022-12-03 | 340            | VIP    |
|   | 209         | Luca       | Esposito  | 555-0109 | luca.esposito@email.com  | Queens        | 2022-09-15 | 520            | VIP    |
| * | NULL        | NULL       | NULL      | NULL     | NULL                     | NULL          | NULL       | NULL           | NULL   |

## Section E: ORDER BY

**Q21. Display all pizzas ordered by price in descending order**

```
SELECT * FROM pizzas
ORDER BY price DESC;
```

103 ORDER BY loyalty\_points DESC;  
186

|  | pizza_id | pizza_name        | size        | crust_type    | price | calories | category   | is_vegetarian |
|--|----------|-------------------|-------------|---------------|-------|----------|------------|---------------|
|  | 5        | Meat Lovers       | Extra Large | Thick Crust   | 22.99 | 450      | Meat       | 0             |
|  | 8        | Spicy Italian     | Large       | Stuffed Crust | 20.99 | 400      | Specialty  | 0             |
|  | 6        | BBQ Chicken       | Large       | Hand Tossed   | 19.99 | 380      | Specialty  | 0             |
|  | 2        | Pepperoni Supreme | Large       | Hand Tossed   | 18.99 | 350      | Meat       | 0             |
|  | 11       | Garden Fresh      | Large       | Thin Crust    | 17.99 | 290      | Vegetarian | 1             |
|  | 10       | Buffalo Chicken   | Medium      | Thick Crust   | 16.99 | 340      | Specialty  | 0             |
|  | 3        | Hawaiian Paradise | Medium      | Stuffed Crust | 15.99 | 320      | Specialty  | 0             |
|  | 7        | Mushroom Magic    | Medium      | Thin Crust    | 13.99 | 260      | Vegetarian | 1             |

**Q22:Show all customers ordered by their loyalty points (highest first).**

#Q22

SELECT \* FROM customers

ORDER BY loyalty\_points DESC;

|   | customer_id | first_name | last_name | phone    | email                    | city          | join_date  | loyalty_points | status  |
|---|-------------|------------|-----------|----------|--------------------------|---------------|------------|----------------|---------|
| ▶ | 205         | Marco      | Lombardi  | 555-0105 | marco.lombardi@email.com | Bronx         | 2022-08-18 | 890            | Gold    |
|   | 203         | Giuseppe   | Bianchi   | 555-0103 | giuseppe.b@email.com     | Manhattan     | 2022-11-10 | 750            | Gold    |
|   | 209         | Luca       | Esposito  | 555-0109 | luca.esposito@email.com  | Queens        | 2022-09-15 | 520            | VIP     |
|   | 201         | Tony       | Soprano   | 555-0101 | tony.soprano@email.com   | Newark        | 2023-01-15 | 450            | VIP     |
|   | 207         | Antonio    | Conti     | 555-0107 | antonio.conti@email.com  | Staten Island | 2022-12-03 | 340            | VIP     |
|   | 202         | Maria      | Rossi     | 555-0102 | maria.rossi@email.com    | Brooklyn      | 2023-02-20 | 280            | Regular |
|   | 208         | Francesca  | Moretti   | 555-0108 | francesca.m@email.com    | Manhattan     | 2023-01-28 | 180            | Regular |
|   | 204         | Isabella   | Ferrari   | 555-0104 | isabella.f@email.com     | Queens        | 2023-03-05 | 120            | Regular |
|   | 206         | Sofia      | Romano    | 555-0106 | sofia.romano@email.com   | Brooklyn      | 2023-04-12 | 65             | Regular |

pizzas 17 customers 18 pizzas 19 orders 20 customers 21 pizzas 22 customers 23 x pizzas 24

**q23:Display pizzas ordered by calories in ascending order.**

SELECT \* FROM pizzas

ORDER BY calories;

|   | pizza_id | pizza_name        | size   | crust_type    | price | calories | category   | is_vegetarian |
|---|----------|-------------------|--------|---------------|-------|----------|------------|---------------|
| ▶ | 4        | Veggie Delight    | Small  | Thin Crust    | 10.99 | 220      | Vegetarian | 1             |
|   | 7        | Mushroom Magic    | Medium | Thin Crust    | 13.99 | 260      | Vegetarian | 1             |
|   | 1        | Margherita        | Medium | Thin Crust    | 12.99 | 280      | Classic    | 1             |
|   | 11       | Garden Fresh      | Large  | Thin Crust    | 17.99 | 290      | Vegetarian | 1             |
|   | 9        | Four Cheese       | Small  | Hand Tossed   | 11.99 | 310      | Classic    | 1             |
|   | 3        | Hawaiian Paradise | Medium | Stuffed Crust | 15.99 | 320      | Specialty  | 0             |
|   | 10       | Buffalo Chicken   | Medium | Thick Crust   | 16.99 | 340      | Specialty  | 0             |
|   | 2        | Pepperoni Supreme | Large  | Hand Tossed   | 18.99 | 350      | Meat       | 0             |
|   | 6        | BBQ Chicken       | Large  | Hand Tossed   | 19.99 | 380      | Specialty  | 0             |

**Q24. Show customers ordered by city name alphabetically, then by last Name.**

```
SELECT * FROM customers
ORDER BY city, last_name;
```

| customer_id | first_name | last_name | phone    | email                    | city      | join_date  | loyalty_points | status   |
|-------------|------------|-----------|----------|--------------------------|-----------|------------|----------------|----------|
| 205         | Marco      | Lombardi  | 555-0105 | marco.lombardi@email.com | Bronx     | 2022-08-18 | 890            | Gold     |
| 210         | Giulia     | Ricci     | 555-0110 | giulia.ricci@email.com   | Bronx     | 2023-05-22 | 15             | Inactive |
| 206         | Sofia      | Romano    | 555-0106 | sofia.romano@email.com   | Brooklyn  | 2023-04-12 | 65             | Regular  |
| 202         | Maria      | Rossi     | 555-0102 | maria.rossi@email.com    | Brooklyn  | 2023-02-20 | 280            | Regular  |
| 203         | Giuseppe   | Bianchi   | 555-0103 | giuseppe.b@email.com     | Manhattan | 2022-11-10 | 750            | Gold     |
| 208         | Francesca  | Moretti   | 555-0108 | francesca.m@email.com    | Manhattan | 2023-01-28 | 180            | Regular  |
| 201         | Tony       | Soprano   | 555-0101 | tony.soprano@email.com   | Newark    | 2023-01-15 | 450            | VIP      |
| 209         | Luca       | Esposito  | 555-0109 | luca.esposito@email.com  | Queens    | 2022-09-15 | 520            | VIP      |
| 204         | Isabella   | Ferrari   | 555-0104 | isabella.f@email.com     | Queens    | 2023-03-05 | 120            | Regular  |

**Q25. Display orders ordered by order\_date (newest first), then by total\_amount (highest first).**

#Q25

```
SELECT * FROM orders
ORDER BY order_date, total_amount DESC;
```

| order_id | customer_id | order_date | delivery_time | total_amount | delivery_type | order_status | special_instructions       |
|----------|-------------|------------|---------------|--------------|---------------|--------------|----------------------------|
| 301      | 201         | 2024-01-15 | 18:30:00      | 45.97        | Delivery      | Delivered    | Extra cheese on all pizzas |
| 303      | 203         | 2024-01-16 | 19:45:00      | 67.95        | Dine-in       | Delivered    | NULL                       |
| 302      | 202         | 2024-01-16 | 12:15:00      | 23.98        | Pickup        | Ready        | No onions please           |
| 304      | 204         | 2024-01-17 | 20:00:00      | 32.99        | Delivery      | Preparing    | Call before delivery       |
| 305      | 205         | 2024-01-17 | 13:30:00      | 18.99        | Pickup        | Ready        | Extra spicy                |
| 307      | 207         | 2024-01-18 | 19:00:00      | 41.97        | Dine-in       | Cancelled    | Table for 4                |
| 306      | 206         | 2024-01-18 | 17:15:00      | 28.98        | Delivery      | Delivered    | NULL                       |
| 309      | 209         | 2024-01-19 | 21:30:00      | 52.96        | Delivery      | Delivered    | Ring doorbell twice        |

## Section F: LIMIT and OFFSET

**Q26. Show the top 3 most expensive pizzas.**

```
SELECT price, pizza_name FROM pizzas
ORDER BY price DESC
LIMIT 3;
```

|   | price | pizza_name     |
|---|-------|----------------|
| ▶ | 24.99 | Supreme Deluxe |
|   | 22.99 | Meat Lovers    |
|   | 20.99 | Spicy Italian  |

**Q27. Display the 5 customers with the most loyalty points.**

```
SELECT first_name, last_name FROM customers
ORDER BY loyalty_points DESC
LIMIT 5;
```

| Result Grid |            |  |  Filter Rows: |
|-------------|------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|             | first_name | last_name                                                                         |                                                                                                |
| ▶           | Marco      | Lombardi                                                                          |                                                                                                |
|             | Giuseppe   | Bianchi                                                                           |                                                                                                |
|             | Luca       | Esposito                                                                          |                                                                                                |
|             | Tony       | Soprano                                                                           |                                                                                                |
|             | Antonio    | Conti                                                                             |                                                                                                |

**Q28. Show pizzas ranked 4-6 when ordered by price (use OFFSET).**

```
SELECT price, pizza_name FROM pizzas
ORDER BY price DESC
LIMIT 3
OFFSET 3;
```

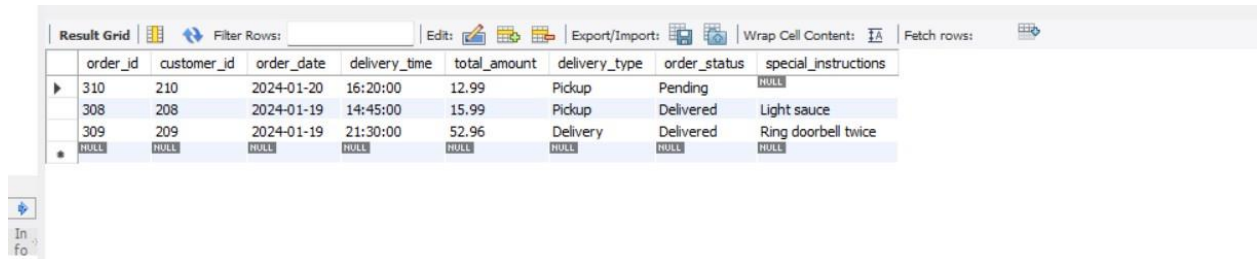
Result Grid

Filter Rows:

|   | price | pizza_name        |
|---|-------|-------------------|
| ▶ | 19.99 | BBQ Chicken       |
|   | 18.99 | Pepperoni Supreme |
|   | 17.99 | Garden Fresh      |

**Q29. Display the 3 most recent orders.**

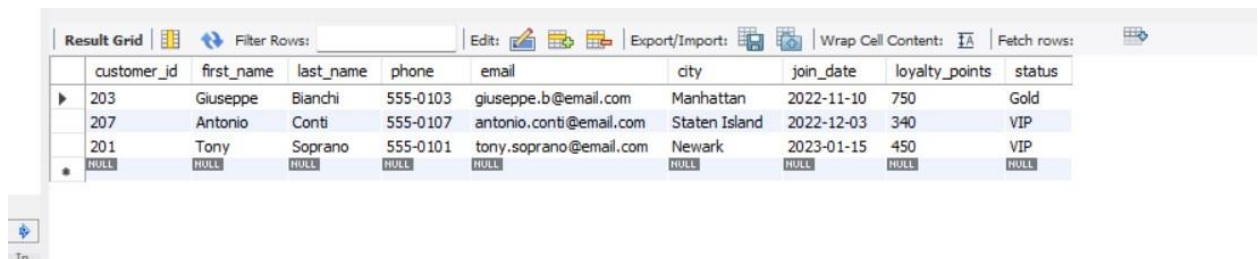
```
SELECT * FROM orders
ORDER BY order_date DESC
limit 3;
```



| order_id | customer_id | order_date | delivery_time | total_amount | delivery_type | order_status | special_instructions |
|----------|-------------|------------|---------------|--------------|---------------|--------------|----------------------|
| 310      | 210         | 2024-01-20 | 16:20:00      | 12.99        | Pickup        | Pending      |                      |
| 308      | 208         | 2024-01-19 | 14:45:00      | 15.99        | Pickup        | Delivered    | Light sauce          |
| 309      | 209         | 2024-01-19 | 21:30:00      | 52.96        | Delivery      | Delivered    | Ring doorbell twice  |

**Q30. Show customers 3-5 when ordered by join date**

```
SELECT * FROM customers
ORDER BY join_date
LIMIT 3
OFFSET 2;
```



| customer_id | first_name | last_name | phone    | email                   | city          | join_date  | loyalty_points | status |
|-------------|------------|-----------|----------|-------------------------|---------------|------------|----------------|--------|
| 203         | Giuseppe   | Bianchi   | 555-0103 | giuseppe.b@email.com    | Manhattan     | 2022-11-10 | 750            | Gold   |
| 207         | Antonio    | Conti     | 555-0107 | antonio.conti@email.com | Staten Island | 2022-12-03 | 340            | VIP    |
| 201         | Tony       | Soprano   | 555-0101 | tony.soprano@email.com  | Newark        | 2023-01-15 | 450            | VIP    |

## Section G: Aliases

**Q31. Display pizza information with aliases: pizza\_name as 'Pizza', size as 'Size', price as 'Cost'.**

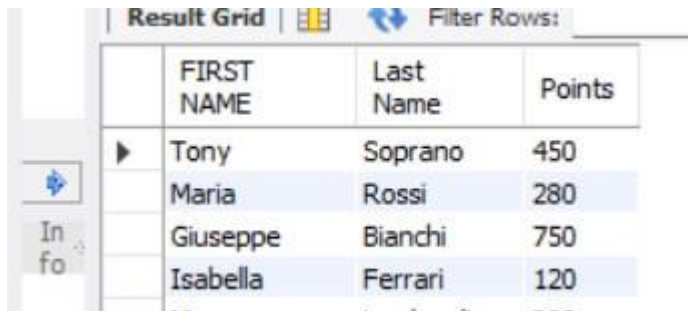
```
SELECT pizza_name AS Pizza, size AS Size, price AS Cost FROM
pizzas;
```



| Pizza             | Size        | Cost  |
|-------------------|-------------|-------|
| Margherita        | Medium      | 12.99 |
| Pepperoni Supreme | Large       | 18.99 |
| Hawaiian Paradise | Medium      | 15.99 |
| Veggie Delight    | Small       | 10.99 |
| Meat Lovers       | Extra Large | 22.99 |

**Q32. Show customer names with aliases: first\_name as 'First Name', last\_name as 'Last Name', loyalty\_points as 'Points'.**

```
SELECT first_name AS 'FIRST NAME', last_name AS 'Last Name', loyalty_points AS Points
FROM customers;
```



The screenshot shows a 'Result Grid' with columns: FIRST NAME, Last Name, and Points. The data rows are:

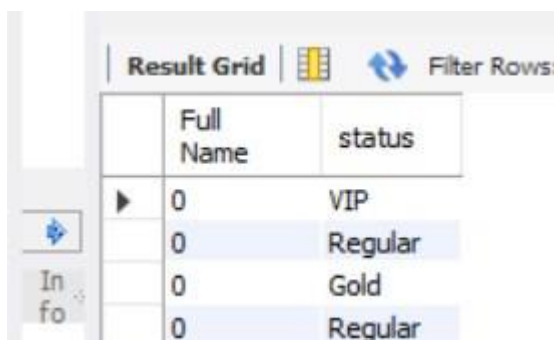
| FIRST NAME | Last Name | Points |
|------------|-----------|--------|
| Tony       | Soprano   | 450    |
| Maria      | Rossi     | 280    |
| Giuseppe   | Bianchi   | 750    |
| Isabella   | Ferrari   | 120    |

**Q33. Create a query showing customer's full name (concatenated) as 'Full Name' and their status as 'Membership Level'.**

```
SELECT first_name+last_name AS 'Full Name', status FROM customers;
```

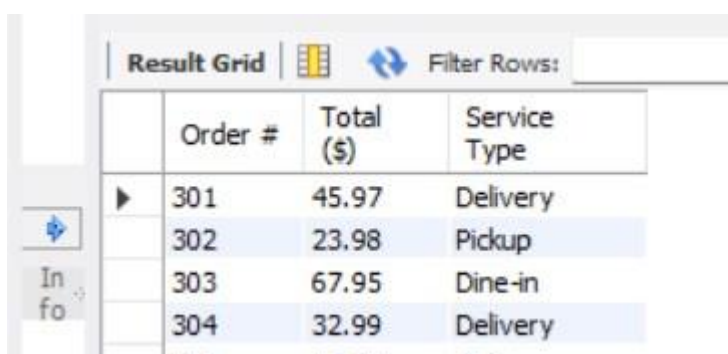
**Q34. Display order information with aliases: order\_id as 'Order #', total\_amount as 'Total (\$)', delivery\_type as 'Service Type'.**

```
SELECT order_id AS 'Order #', total_amount as 'Total ($)', delivery_type as 'Service Type' FROM orders;
```



The screenshot shows a 'Result Grid' with columns: Full Name and status. The data rows are:

| Full Name | status  |
|-----------|---------|
| 0         | VIP     |
| 0         | Regular |
| 0         | Gold    |
| 0         | Regular |



The screenshot shows a 'Result Grid' with columns: Order #, Total (\$), and Service Type. The data rows are:

| Order # | Total (\$) | Service Type |
|---------|------------|--------------|
| 301     | 45.97      | Delivery     |
| 302     | 23.98      | Pickup       |
| 303     | 67.95      | Dine-in      |
| 304     | 32.99      | Delivery     |

## Section H: Combined Operations

**Q35. Find the top 3 most expensive vegetarian pizzas.**

```
SELECT price, pizza_name FROM pizzas
WHERE is_vegetarian=1
ORDER BY price DESC
LIMIT 3;
```

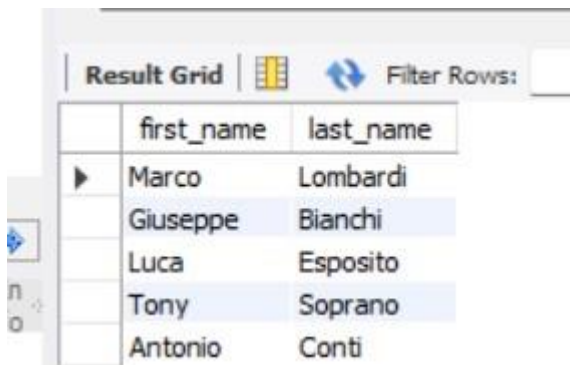


The screenshot shows a database query result grid with two columns: 'price' and 'pizza\_name'. The results are ordered by price in descending order. The first row is 'Garden Fresh' with a price of 17.99. The second row is 'Mushroom Magic' with a price of 13.99. The third row is 'Margherita' with a price of 12.99.

| price | pizza_name     |
|-------|----------------|
| 17.99 | Garden Fresh   |
| 13.99 | Mushroom Magic |
| 12.99 | Margherita     |

**Q36. Display the first 5 VIP customers ordered by loyalty points (highest first).**

```
SELECT first_name, last_name FROM customers
ORDER BY loyalty_points DESC
LIMIT 5;
```

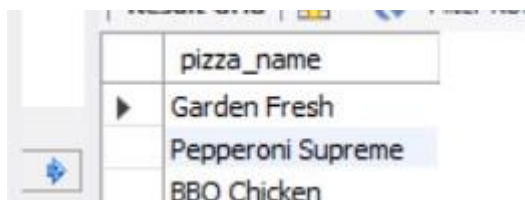


The screenshot shows a database query result grid with two columns: 'first\_name' and 'last\_name'. The results are ordered by loyalty points in descending order. The first row is 'Marco Lombardi'. The second row is 'Giuseppe Bianchi'. The third row is 'Luca Esposito'. The fourth row is 'Tony Soprano'. The fifth row is 'Antonio Conti'.

| first_name | last_name |
|------------|-----------|
| Marco      | Lombardi  |
| Giuseppe   | Bianchi   |
| Luca       | Esposito  |
| Tony       | Soprano   |
| Antonio    | Conti     |

**Q37. Show Large pizzas under \$20, ordered by calories, limit to 3 results.**

```
SELECT pizza_name FROM pizzas
WHERE size="Large" AND price<20.00
ORDER BY calories
LIMIT 3;
```



The screenshot shows a database query result grid with one column: 'pizza\_name'. The results are ordered by calories. The first row is 'Garden Fresh'. The second row is 'Pepperoni Supreme'. The third row is 'BBQ Chicken'.

| pizza_name        |
|-------------------|
| Garden Fresh      |
| Pepperoni Supreme |
| BBQ Chicken       |

**Q38. Find customers from Brooklyn or Queens with more than 200 loyalty points, ordered by points.**

```
SELECT first_name, last_name FROM customers
WHERE (city="Brooklyn" OR city="Queens") AND loyalty_points>200
ORDER BY loyalty_points;
```



The screenshot shows a database interface with a 'Result Grid' tab. It displays a table with two columns: 'first\_name' and 'last\_name'. There are two rows of data: 'Maria Rossi' and 'Luca Esposito'. The second row is highlighted. To the left of the table is a vertical toolbar with icons for navigation and filtering. Above the table is a 'Filter Rows:' input field.

|   | first_name | last_name |
|---|------------|-----------|
| ▶ | Maria      | Rossi     |
|   | Luca       | Esposito  |

**Q39. Display the 2nd and 3rd cheapest pizzas in the 'Specialty' category.**

```
SELECT pizza_name FROM pizzas
WHERE category="Specialty"
ORDER BY price
LIMIT 2
OFFSET 1;
```



The screenshot shows a database interface with a 'Result Grid' tab. It displays a table with one column: 'pizza\_name'. There are two rows of data: 'Buffalo Chicken' and 'BBQ Chicken'. The second row is highlighted. To the left of the table is a vertical toolbar with icons for navigation and filtering.

|   | pizza_name      |
|---|-----------------|
| ▶ | Buffalo Chicken |
|   | BBQ Chicken     |

**Q40. Show orders from January 17-19, 2024, with delivery type 'Delivery', ordered by total amount.**

```
SELECT * FROM orders
WHERE order_date BETWEEN "2024-01-17" AND "2024-01-19"
AND delivery_type="Delivery"
ORDER BY total_amount;
```

|                    |          |             |            |               |              |               |              |                      |
|--------------------|----------|-------------|------------|---------------|--------------|---------------|--------------|----------------------|
| Result Grid        |          |             |            |               |              |               |              |                      |
| Filter Rows:       |          |             |            |               |              |               |              |                      |
| Edit:              |          |             |            |               |              |               |              |                      |
| Export/Import:     |          |             |            |               |              |               |              |                      |
| Wrap Cell Content: |          |             |            |               |              |               |              |                      |
|                    | order_id | customer_id | order_date | delivery_time | total_amount | delivery_type | order_status | special_instructions |
| ▶                  | 306      | 206         | 2024-01-18 | 17:15:00      | 28.98        | Delivery      | Delivered    | NULL                 |
|                    | 304      | 204         | 2024-01-17 | 20:00:00      | 32.99        | Delivery      | Preparing    | Call before delivery |
|                    | 309      | 209         | 2024-01-19 | 21:30:00      | 52.96        | Delivery      | Delivered    | Ring doorbell twice  |